



Setter And Getter

By Kunal Sir

Assignments

What are Setters and Getters?

- **Getters:** Methods used to retrieve the value of a private field (variable) in a class.
- **Setters:** Methods used to modify or set the value of a private field in a class.

Rules:

- 1) The class must be public
- 2) make the variable (field) as private
- 3) The setter method is declared as public, takes a parameter, and does not return any value
- 4) The getter method is declared as public, does not take any parameters, and returns a value.

Why we use?

Setter and getter methods in Java help protect class fields by controlling how private data is accessed and changed. Setters check if the input values are valid, while getters allow you to get the values of the fields.

How to Define Setters and Getters

Example:



```
public class A { // declare class as public
```

```
    // to make fields as private
```

```
    private int a;
```

```
    // public Getter method for name
```

```
    public int getA() {
```

```
        return a;
```

```
    }
```

```
    // public Setter method for name
```

```
    public void setA(int a) {
```

```
        this.a = a;
```

```
    }
```

```
    }
```

```
}
```

```
public class Test {
```

```
    public static void main(String[] args) {
```



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By Kunal Sir

// Create an object of the A class

```
A a = new A();
```

// Set values using setters

```
a.setA(10);
```

// Get values using getters

```
System.out.println("A : " + a.getA());
```

```
}
```

```
}
```

Output:

A:10

Assignment 1: Basic Student Class

Create a Student class with the following private fields:

- name (String)
- rollNumber (int)



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By Kunal Sir

- grade (char)

Write appropriate setters and getters for these fields

Write a main method to test your class by creating a Student object, setting its fields, and displaying the values.

Assignment 2: Bank Account Class

Create a BankAccount class with the following private fields:

- accountNumber (String)
- accountHolderName (String)
- balance (double)

Write appropriate setters and getters for these fields.

Write a main method to test your class by creating a BankAccount object, setting its fields, and displaying the values.

Assignment 3: Employee Class

Create an Employee class with the following private fields:

- employeeId (int)
- employeeName (String)
- salary (double)
- department (String)



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Write appropriate setters and getters for these fields

Write a main method to test your class by creating an Employee object, setting its fields, and displaying the values.

Assignment 4: Book Class

Create a Book class with the following private fields:

- title (String)
- author (String)
- isbn (String) (this field should be read-only after initialization)
- price (double)

Write appropriate setters and getters for these fields.

Write a main method to test your class by creating a Book object, setting its fields, and displaying the values.

Assignment 5: Car Class

Create a Car class with the following private fields:

- make (String)
- model (String)
- year (int)
- mileage (double)



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By Kunal Sir

Write appropriate setters and getters for these fields.

Write a main method to test your class by creating a Book object, setting its fields, and displaying the values.